

## DEPARTMENT OF INFORMATION TECHNOLOGY

### Subject wise Common Question Bank (QB) for All Divisions

|                                                                                                                                         |                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Class:</b>                                                                                                                           | SY/TY/BTech                                  |
| <b>Subject Name/Code:</b>                                                                                                               | Data Structure and Algorithms (UITL203)      |
| <b>Subject Teacher Name(s):</b>                                                                                                         | 1.Mr. Yogesh Gite<br>2. Mrs. Sheetal Borhade |
| <b>Note:</b> Only 60% Questions can be asked from this QB in CAE. Rest of 40% questions will be from out of this QB related to syllabus |                                              |

| Unit-I  |                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q. No.  | Question Statement                                                                                                                                                                                                                                                                                                                                                                          |
| 1.      |                                                                                                                                                                                                                                                                                                                                                                                             |
| 2.      | What is a data structure? Explain different types of Data structures.                                                                                                                                                                                                                                                                                                                       |
| 3.      | Compare Linear and Non-Linear Data Structure.                                                                                                                                                                                                                                                                                                                                               |
| 4.      | Elaborate Array as ADT.                                                                                                                                                                                                                                                                                                                                                                     |
| 5.      | What is time and space complexity of an algorithm. Give necessary example.                                                                                                                                                                                                                                                                                                                  |
| 6.      | Define algorithm. Write an algorithm for linear search.                                                                                                                                                                                                                                                                                                                                     |
| 7.      | Explain with neat diagram asymptotic notations.                                                                                                                                                                                                                                                                                                                                             |
| 8.      | Write a C program to perform linear search on an array.                                                                                                                                                                                                                                                                                                                                     |
| 9.      | Write a C program to delete an element from an array.                                                                                                                                                                                                                                                                                                                                       |
| 10.     | Find the time complexity of following code.<br><div style="border: 1px solid #ccc; padding: 10px; margin: 10px 0;"> <p><b>C Code:</b></p> <pre>#include &lt;stdio.h&gt;  int main() {     int n, sum = 0;     scanf("%d", &amp;n);     for (int i = 1; i &lt;= n; i++) {         sum += i;     }     printf("Sum of first %d natural numbers = %d\n", n, sum);     return 0; }</pre> </div> |
| Unit-II |                                                                                                                                                                                                                                                                                                                                                                                             |
| Q. No.  | Question Statement                                                                                                                                                                                                                                                                                                                                                                          |
| 1.      | What is sorting? List the different types of sorting techniques.                                                                                                                                                                                                                                                                                                                            |
| 2.      | Write bubble sort algorithm. Sort the elements of array using bubble sort.<br>5,2,8,4,15,12,7                                                                                                                                                                                                                                                                                               |
| 3.      | Write selection sort algorithm. Sort the elements of array using bubble sort.<br>5,2,8,4,15,12,7                                                                                                                                                                                                                                                                                            |
| 4.      | Write insertion sort algorithm. Sort the elements of array using bubble sort.                                                                                                                                                                                                                                                                                                               |

|            |                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------|
|            | <b>5,2,8,4,15,12,7</b>                                                                                                |
| <b>5.</b>  | <b>Define hashing. Write types of hashing and give examples of hashing.</b>                                           |
| <b>6.</b>  | <b>Compare Linear and Binary Search</b>                                                                               |
| <b>7.</b>  | <b>What are different collision avoidance and collision resolution techniques in hashing.</b>                         |
| <b>8.</b>  | <b>Define Linked List. Give types of linked lists.</b>                                                                |
| <b>9.</b>  | <b>List the operations performed on linked lists. Write pseudo code for inserting an element in singly link list.</b> |
| <b>10.</b> | <b>Write a pseudo code to create and display elements of singly link list</b>                                         |